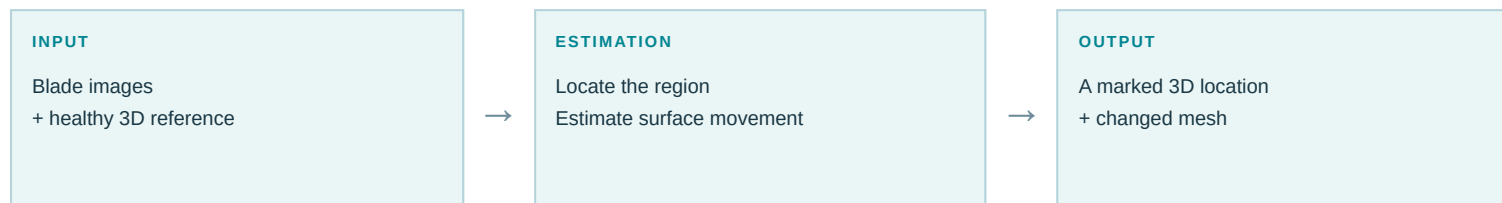
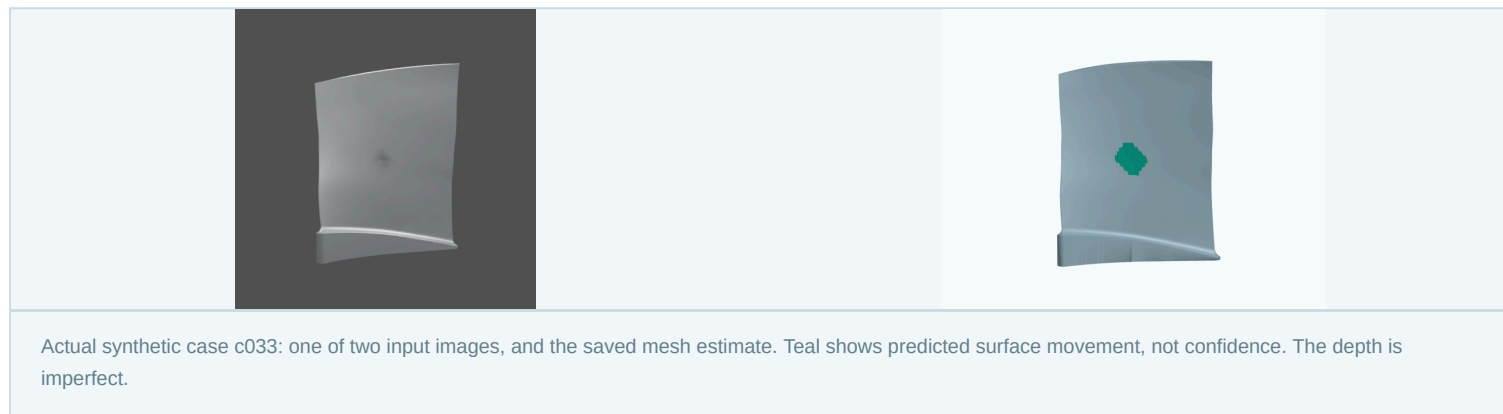


From blade images to a 3D damage estimate.

Can inspection images tell us where a blade physically changed—and recover the actual change in its three-dimensional shape?



Both the location and the estimated shape are in 3D. The healthy blade is supplied in our current experiments. This is not a claim that we can recover an unknown blade accurately from an arbitrary photograph.



What has already been built?

1. Locate a detection on known 3D geometry.

A learned image detector selects a point. Known camera and surface records map it onto a supplied mesh. A correct point does not guarantee a complete damage mask.

2. Learn to predict surface movement.

Small learned models use a healthy mesh and paired image evidence to produce actual changed-mesh outputs. False movements and missed damage remain problems.

3. Investigate the errors in controlled cases.

The latest experiment compares shape-only and shape-plus-color fitting. It is a restricted, nonlearned diagnostic—not the final proposed method.

A working prototype.

An open research question.

The latest test retains **68 cases from two exposed source shapes**: 18 dents, 12 color-only changes, 36 mixtures and 2 healthy controls. Both fits use the same two input images, known cameras and lighting, and three possible dent locations.

99.8% less

False shape movement

Separating color from shape reduced mean false-displacement RMS across the 12 color-only cases.

26.8% higher

Local shape error

Mean local RMS error increased on mixed cases. The newer fit worsened 30 of 36 examples.

Joint versus shape-only fitting. Percentages compare means of per-case RMS, not accuracy. Local error uses exactly changed blade-surface vertices. Complete results and controls remain available.

The lesson: a method can find the right supplied location and match the images better, yet estimate a worse 3D shape. Reliable geometry remains unproven.

What happens next?

1. Render the estimated meshes and materials directly.

Check whether the current image approximation explains the depth errors. This is the next planned experiment; it has not been launched.

2. Improve the estimator and compare fairly.

Measure actual 3D accuracy, missed damage and false surface movement. Compare with simpler methods using the same inputs.

3. Freeze the method, then test unfamiliar conditions.

Use untouched shapes, new damage locations and changes in cameras or lighting. Avoid selecting only favorable examples.

4. Validate on independently measured real damage.

Establish physical accuracy and uncertainty before claiming reliable real-world inspection.

The intended final output

A marked surface, estimated changed shape, linked supporting images and calibrated uncertainty. If evidence is insufficient, request a useful additional view.

A related research direction

Use known 3D damage during learning to improve real-photo damage masks. This is a separate unproven question, not a result of the current fit.

Automatic next-view selection, trustworthy uncertainty and real-blade validation are future work. No completion percentage, safety decision, physical millimeter accuracy or invented confidence score is claimed.